35 Agriculture and subsistence in mountain sites: a FLOSS approach to the reconstruction of the subsistence economy of the Bronze Age settlement of Monte della Croce (Bologna, Italy)

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This paper presents a multidisciplinary approach for creating a spatial model aimed at reconstructing the subsistence strategies and the exploitation of plant resources at a mountain site during the Late Bronze Age, specifically Monte della Croce (Upper Reno Valley, 922 m a.s.l.).

The research integrates quantitative and statistical data obtained from archaeobotanical analyses of five samples with spatial analysis, land evaluation, and carrying capacity models. The study was conducted using Free/Libre and Open Source Software (FLOSS) solutions, developing a spatial analysis model (land suitability) based on the calculation of the territory controlled by the site of Monte della Croce and its potential agricultural capability, implemented within QGIS. The goal is to produce a realistic simulation of land use and management, subsequently calculating from it the number of sustainable individuals (carrying capacity) based on the yields of ancient cereals and the average caloric requirement per person. According to the calculation, the area dedicated to cereal cultivation is 592 hectares, representing 67.5% of the total estimated controlled territory.

Specifically, frequency data of cultivated and collected species were integrated with Digital Elevation Models (DEMs) and spatial analysis tools, critical environmental variables such as slope, solar radiation, and the topographic wetness index, as well as agronomic criteria relevant to the study area and the adopted subsistence economy. Algorithms were applied to estimate the carrying capacity and, consequently, the demographic potential of the settlement—an estimation difficult to obtain for high-altitude settlements, where extensive excavations are rare.